

How to run my code

The code for this exercise was implemented in Python using the PySAT library. The graph input files should be available in the same working directory as the Python code.

Before running the code, install the required PySAT package:

```
pip install python-sat
```

The code can then be run using Python or through a Python environment such as Jupyter Notebook.

For **Problem 1**, the program reads a graph from an input file and takes the number of colours k as input. It constructs the corresponding CNF formula and uses the Glucose3 SAT solver to determine whether the graph can be coloured using at most k colours. The program returns True if the graph is colourable and False otherwise.

For **Problem 2**, the program reads the graph and uses three possible states for each vertex: red, green, or removed. The decision solver is tested with increasing values of k , starting from $k = 0$. The first value of k for which the problem is satisfiable gives the minimum number of vertices that must be removed. The corresponding set of removed vertices W is then extracted from the satisfying assignment.

The experiments use the graph files described in the report, including graph.txt, path.txt, two_triangle.txt, larger.txt, large_10.txt, large_20.txt, large_30.txt, and large_40.txt.

Runtime is measured using Python's `time.perf_counter()`.